

# Modelling long-term age dynamics of malaria vaccine impact in endemic African settings

## Modelling the long-term impact of a pre-erythrocytic malaria vaccine on endemic settings in Sub-Saharan Africa

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### Introduction

- The impact of malaria vaccination varies depending on local transmission level, intervention coverage, time period, and population demography.
- Mathematical modelling can quantify vaccine impact, informing decision-making surrounding vaccine allocation and campaign strategy.
- Delayed malaria** is characterized by resurgent malaria in people who did not develop natural immunity as children. Delayed malaria holds the potential to undercut vaccine impact by shifting cases to older ages.
- We aimed to model vaccine impact on malaria burden, assessing long-term age dynamics following vaccine introduction in a high-transmission African setting.

### Methodology

- As a part of ongoing work with the Vaccine Impact Modelling Consortium (VIMC), we modelled the impact of vaccine introduction on malaria incidence and mortality in a high-transmission African setting.
- We utilised **malariasimulation**, an Imperial College individual-based model of malaria transmission that has been fitted to epidemiological data from varied transmission settings<sup>1</sup>.
- We ran transmission models with and without pre-erythrocytic vaccines, modelling an age-based vaccination scenario that achieved 80% coverage over a 100-year period. We then compared incidence and mortality rates in the modelled population over this time period.

### Results

- Our models demonstrated that vaccine impact is concentrated in children under five.
- Given that pre-erythrocytic vaccines are partially effective, models predict a long-term shift in cases to older ages due to reduced acquisition of natural immunity in areas of higher transmission.
- Under current assumptions there is less age-shift in deaths averted compared to clinical cases, as we assume higher susceptibility to severe malaria at young ages<sup>2</sup>.
- However, there is limited long-term data in vaccinated children against which to validate these findings.

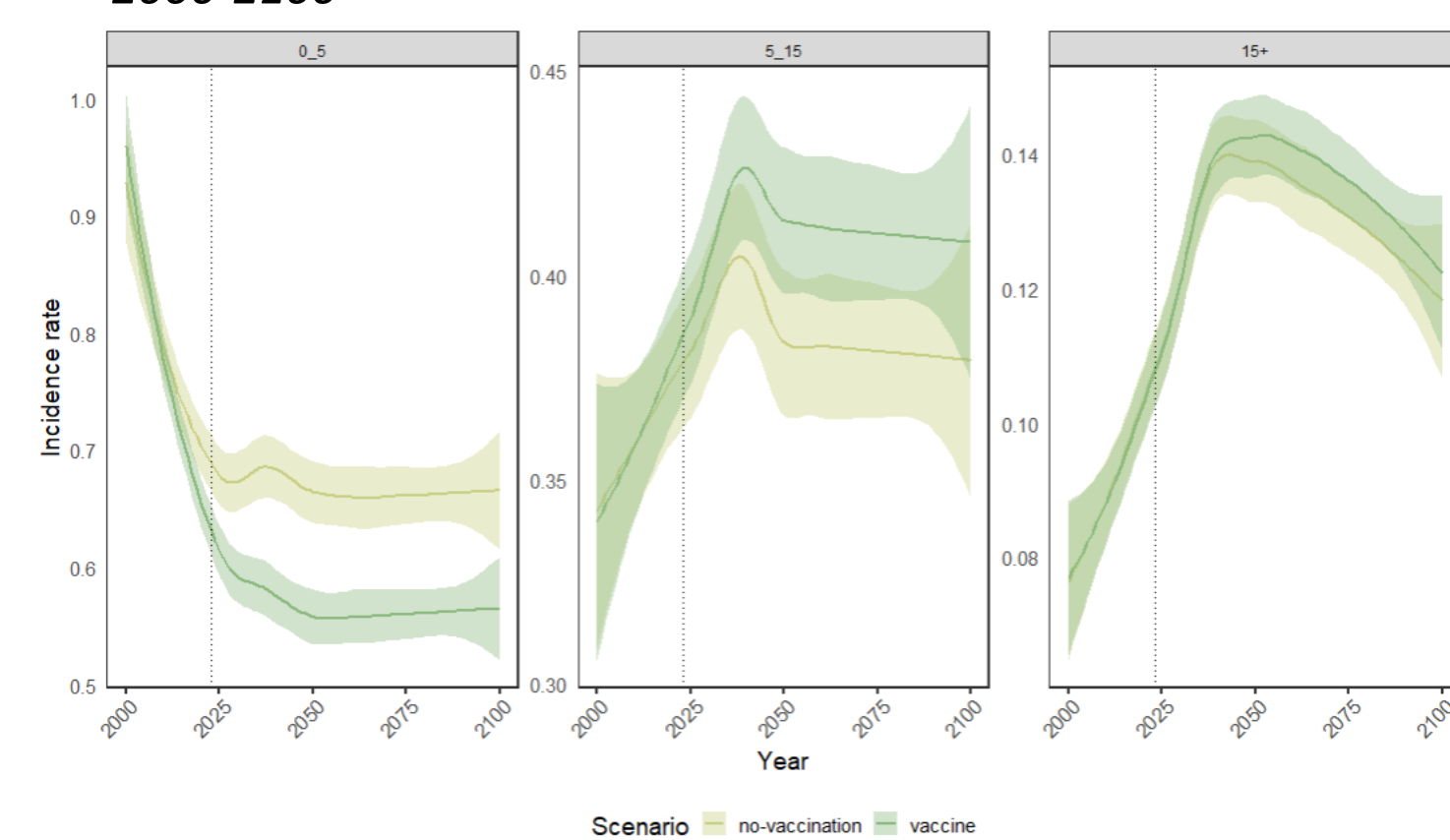
### Discussion & Future Work

- Efforts to model the impact of vaccination are complicated by the dynamic nature of malaria transmission and uncertainties over the duration of protection from vaccination.
- More long-term trial data on vaccine efficacy and duration is needed to reduce uncertainty surrounding vaccine impact estimates, as well as to increase understanding of the long-term dynamics of malarial immunity in children given the vaccine.

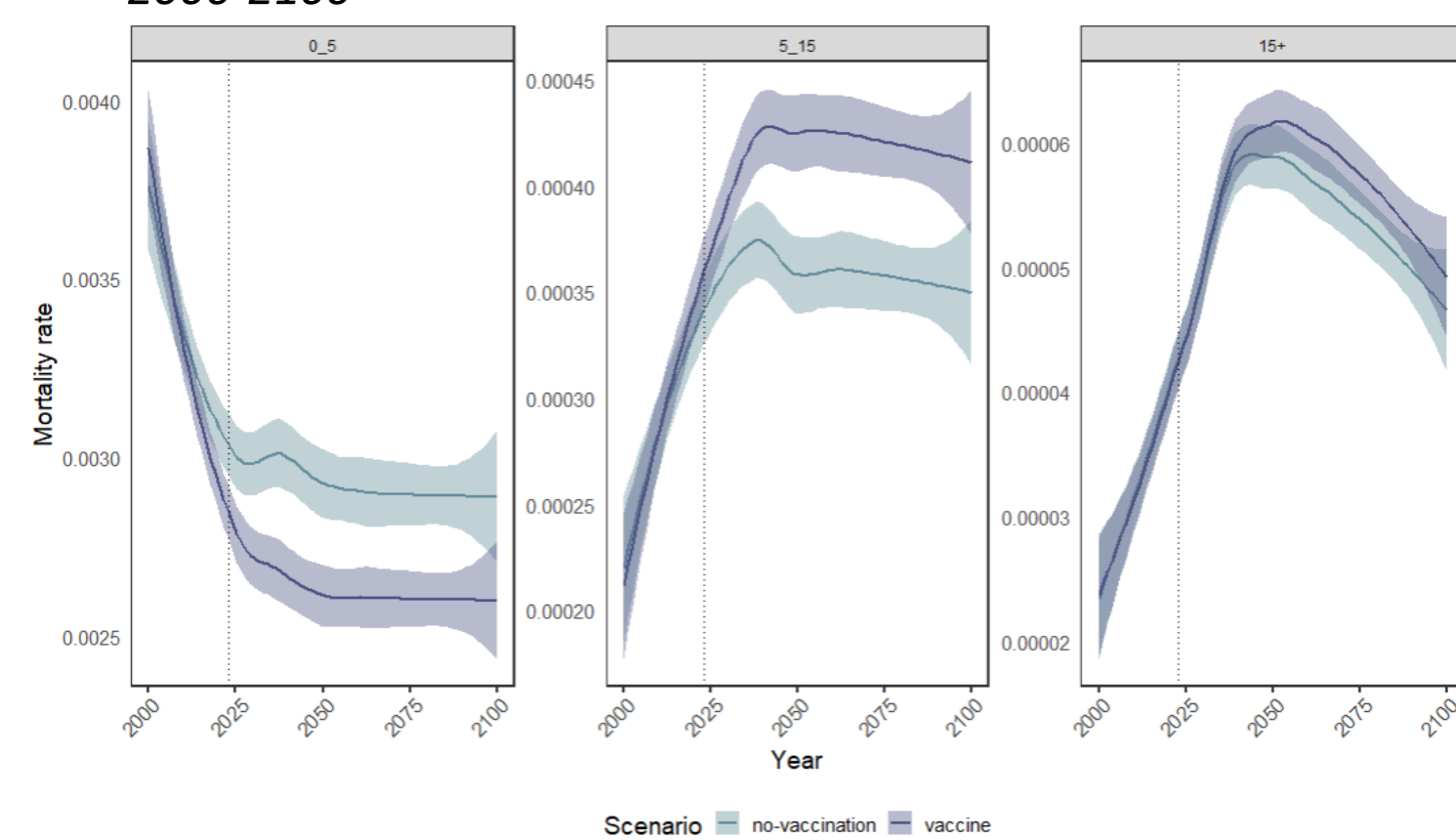
### References

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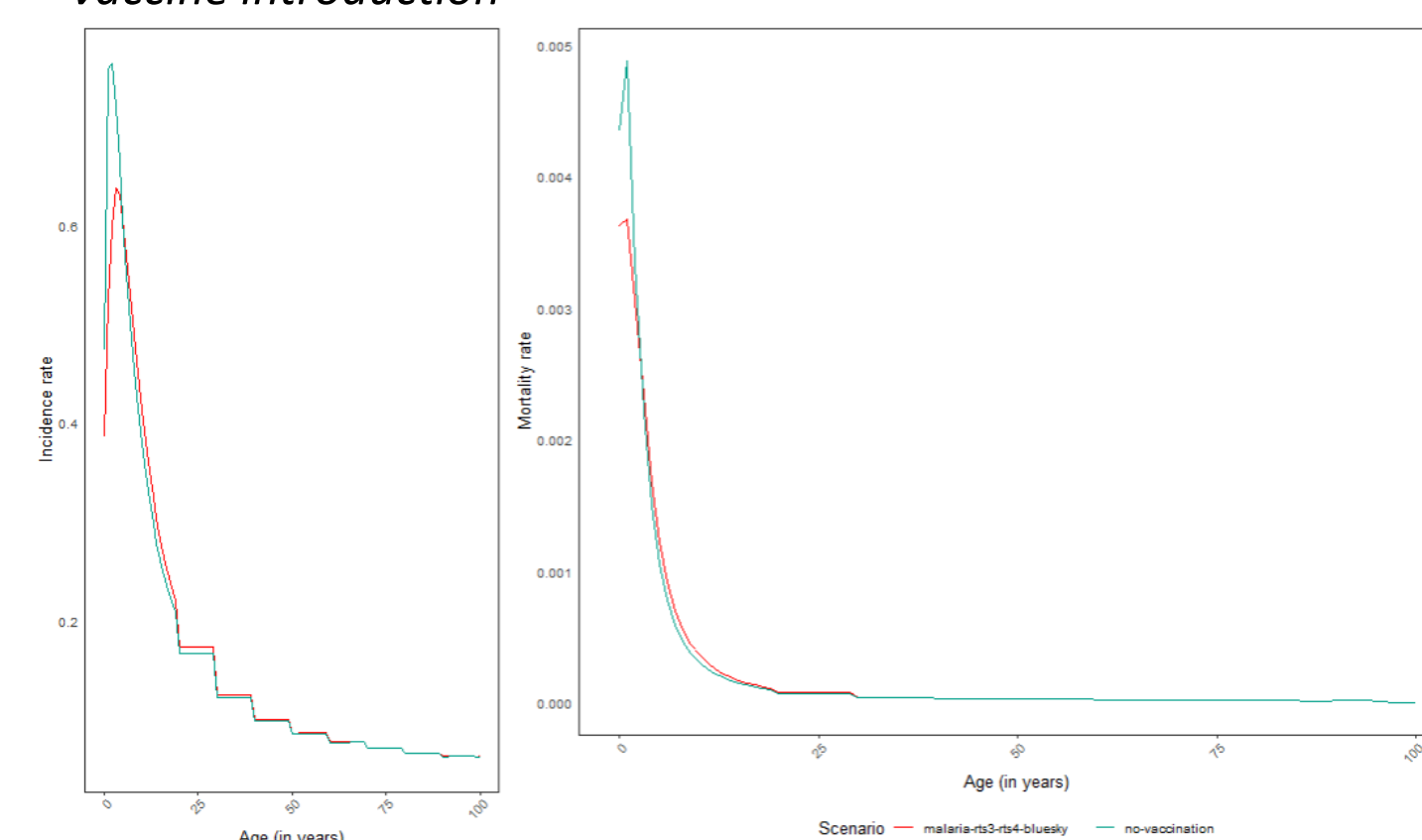
**Figure 1.** Incidence rate by age group and vaccine scenario, 2000-2100



**Figure 2:** Mortality rate by age group and vaccine scenario, 2000-2100



**Figure 3:** Incidence and mortality rate by age 15 years after vaccine introduction



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**Model**

